

DPEN102 ↔ Jeff Hanson "How to Ace Statics" Cross-Reference

Source: *How to Ace Statics with Jeff Hanson* (Hanson, Bernal, Pitarresi, McGraw-Hill 2022) — 9 "Levels," 71 core video lessons on YouTube, each level ending with a "Key Takeaways" summary. Lesson numbers marked (**confirmed**) come directly from the publisher's table of contents; numbers marked (*approx.*) are reconstructed from partial listings and may be off by a lesson or two — check the playlist directly if exact numbers matter.

Quick lookup table

DPEN102 lesson	Hanson Level	Lesson range	Confidence
FBD (standalone)	Level 2 + Level 4	14-20, 29-31	confirmed
Week 1 — Newton's Laws & Equilibrium	Level 1 (start) + Level 2 (start)	1, 14	confirmed
Week 2 — Units, Method & Moments	Level 1 (intro) + Level 3	1, 21-23	confirmed
Week 3 — Vectors, Components & Equilibrium	Level 1 + Level 2 + Level 4	2-13, 14-20, 29	confirmed
Week 4 — Stress, Strain & Internal Actions	partial match: Level 7 only	57--62	approx. (stress/strain not covered by Hanson)
Week 5 — SFD/BMD	Level 7	~58--62	approx.
Week 6 — Truss Method of Sections	Level 6	~47-56	approx. (52 confirmed)
Week 7 — Friction	Level 8	~63-66	approx.
Week 8 — Centroids & Inertia	Level 5 + Level 9	36-46, 67-70	confirmed (endpoints)
Week 9 — Work, Energy & Conservation	no match	—	not in Hanson's Statics course
Week 10 — Integrated Statics Practice	Test Yourself Solutions / Practice Exams	back matter	confirmed (exists)
Week 11 — Design Blocks (Stress/Centroid/I)	partial match: Level 9 only	67-70	approx. (stress-check content not covered)

DPEN102 lesson	Hanson Level	Lesson range	Confidence
Week 12 — Exam-Style Synthesis	Practice Exams / Exam Solutions	back matter	confirmed (exists)
Week 13 — Final Readiness & Closure	Key Takeaways (all levels) + Lesson 71	71	confirmed

Detailed mapping

FBD lesson → Level 2: Particle Equilibrium + Level 4: Global Equilibrium Reactions

- **Lesson 18** — *2D Statics on a Particle, Multiple Free Body Diagrams* is the closest single-lesson match to the whole FBD lesson's spirit.
- **Lessons 14-17** (scalar equations, pulleys, springs) match FBD Sections 1-4 (block on a surface, angled ropes, inclines, connected bodies).
- **Lesson 29** — *2D Reaction at Supports, Example Problem* matches FBD Sections 5-6 (beam supports, ladder/multi-surface problems).

Week 1 (Newton's Laws) → Level 1, Lesson 1

- **Lesson 1** — *Introduction and Newton's Laws, Scalars, and Vectors* covers this week's Sections 1-3 directly (three laws, static equilibrium). Section 4 (dynamics, ($\Sigma F=ma$)) is mentioned in Lesson 1 but not developed further, Hanson's course stays almost entirely in the static ($a=0$) case after this point, consistent with it being a Statics-only series.

Week 2 (Units, Method, Moments) → Level 1 intro + Level 3

- Units, significant figures, and the 6-step method aren't a dedicated Hanson lesson, they're closer to his book's front-matter "Introduction" chapter than a numbered video.
- **Lessons 21-23** — *Introduction to Moments and Torque, 2D Moments About a Point, Two Methods, 2D Min/Max Moment Calculations* map directly to this week's Section 4.

Week 3 (Vectors, Components, Equilibrium) → Level 1 + Level 2 + Level 4

- **Lessons 2-7** — vector addition, the triangle rule, Cartesian/component notation: matches Sections 1-2 closely.
- **Lesson 6** — *Most Missed Topic in Statics, Cartesian Coordinates* is worth watching specifically alongside this week's Section 1 (magnitude-angle ↔ components), Hanson calls out this exact conversion as the single most common source of errors.
- **Lessons 14, 18** — equilibrium equation setup with multiple FBDs, matches Section 4's advanced beam+reactions example.
- (Lessons 8-13, 3D vectors and the dot product, go beyond what DPEN102's Week 3 covers, 2D only.)

Week 4 (Stress/Strain/Internal Actions) → partial: Level 7 only

- **Sections 1–2 (stress, strain, allowable-stress design) have no equivalent in Hanson's course.** This is Mechanics of Materials content; a pure Statics course (which is what Hanson teaches) stops at finding forces and moments, it doesn't get into whether the material survives them.
- **Section 3 (internal actions N , V , M at a cut) matches Lesson 57 — *Introduction to Internal Forces*, M N V almost exactly, same cut-and-isolate-one-side method, same three unknowns.**

Week 5 (SFD/BMD) → Level 7 (later lessons)

- Follows directly after Lesson 57 within Level 7; Hanson's later Level 7 lessons build the shear and moment diagrams the same way, piecewise for point loads, then distributed loads via ($dV/dx=w$), ($dM/dx=V$). Exact lesson numbers weren't confirmed in this search, expect them somewhere around 58–62.

Week 6 (Truss Method of Sections) → Level 6: Trusses, Frames, and Machines

- **Lesson 52 — *Truss Problem, The Combo Problem*** is confirmed and is a strong match for this week's Section 3 (the full method-of-sections algorithm).
- Zero-force members (Section 1) and global truss reactions (Section 2) would be earlier lessons within Level 6, likely in the 47–51 range.

Week 7 (Friction) → Level 8: Friction

- Confirmed to sit between Level 7 (ending somewhere in the low-to-mid 60s) and Level 9 (starting at Lesson 67), so Level 8 likely spans roughly Lessons 63–66.
- The intro lesson description ("introduces friction as an additional force and demonstrates how to find the direction in which it acts") matches this week's Section 0 (the three friction states) directly.

Week 8 (Centroids & Inertia) → Level 5 + Level 9 (two separate levels in Hanson's course)

- **Lessons 42–46 (confirmed): *Introduction to Centroid by Calculus Method, Centroids by Calculus Examples, ..., Distributed Loads Using Centroids*.** Matches this week's Sections 0–1 (composite centroids, addition and subtraction), though Hanson also covers the calculus method for centroids, which DPEN102 doesn't include.
- **Lessons 67–70 (confirmed): *Introduction to Moment of Inertia, Parallel Axis Theorem, Composite Shape Method, Calculus Method*.** Matches this week's Sections 2–3 almost lesson-for-lesson, Hanson even uses "composite shape method" as the name for exactly the workflow in DPEN102 Section 3.
- Worth noting: DPEN102 teaches centroids and inertia in the same week, Hanson splits them into Level 5 and Level 9, with trusses, internal forces, and friction in between. If following Hanson's order, second moment of area comes right at the *end* of the course, not paired with centroids.

Week 9 (Work, Energy, Conservation) → no match

- Not covered anywhere in Hanson's Statics series. Work-energy methods belong to Dynamics, a separate course (and a separate Hanson series, if one exists) — his book's table of contents has no energy-related level at all.

Week 10 (Integrated Statics Practice) → Test Yourself Solutions / Practice Exams (back matter)

- The book's back matter includes "Test Yourself Solutions," "Practice Exams," and "Exam Solutions" sections, exactly the multi-topic, worked-solution format this week uses. No specific lesson numbers apply since this is reference/practice material rather than numbered video content.

Week 11 (Design Blocks) → partial: Level 9 only

- The centroid/second-moment portions reuse Lessons 67–70 (same as Week 8).
- **The allowable-stress design-check portions (Sections 1-2, "is it safe," "max safe load") have no Hanson equivalent**, same gap as Week 4, this is Mechanics of Materials, not Statics.

Week 12 (Exam-Style Synthesis) → Practice Exams / Exam Solutions (back matter)

- Direct structural match: Hanson's book literally includes full practice exams with worked solutions as its closing sections, the same "timed mixed paper" format this week uses.

Week 13 (Final Readiness) → Key Takeaways sections + Lesson 71

- Every one of Hanson's 9 Levels ends with a "Key Takeaways" summary page, exactly the compact formula-reference-sheet idea in this week's Section 1.
- **Lesson 71** — *Statics Conclusion and Farewell! What's Next?* is the literal final video in the series, a natural pairing with this week's closing/readiness framing.

Summary: where DPEN102 and Hanson diverge

DPEN102 covers that Hanson doesn't:

- Stress, strain, allowable-stress design (Weeks 4 §1-2, 11 §1-2) — Mechanics of Materials territory
- Work, energy, conservation (Week 9 entirely) — Dynamics territory

Hanson covers that DPEN102 doesn't:

- Full 3D vector mechanics, moments about an axis, and moment of a couple in 3D (Level 1 §8-13, Level 3 §24-26)
- The calculus method for centroids and second moment of area (Level 5 §42-43, Level 9 §70), DPEN102 stays entirely in the composite-shape (algebraic) method
- Frames and machines specifically (multi-body, non-truss structures) within Level 6

If useful, Hanson's course is the stronger resource for the two DPEN102 gaps' *surrounding* material (pure statics is deeper and more 3D in his series), while DPEN102's Weeks 4 and 11 fill in the stress/design side his course explicitly leaves for a follow-on Mechanics of Materials class.